Project 11

Automated User Onboarding and Offboarding

PowerShell Automation for Active Directory User Lifecycle Management

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Domain	corp.local (NetBIOS: CORP)
Language	PowerShell
Scheduling	Windows Task Scheduler (Hourly)
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1. Project Overview

This project automates the most time-consuming routine task in IT administration: user account lifecycle management. A PowerShell script reads CSV files submitted by HR, automatically creates Active Directory accounts with correct OU placement, security group membership, home folders, and auto-generated credentials for new hires. For departing employees, it disables accounts, removes group memberships, and moves them to a Disabled Users OU. The entire process runs on an hourly schedule via Windows Task Scheduler.

Feature	Manual Process	Automated Process
Create AD account	5-10 minutes	2 seconds
Add to security group	2 minutes	Automatic
Create home folder	3 minutes	Automatic
Generate credentials	5 minutes	Auto-generated random password
Disable departing user	5-10 minutes	2 seconds
Remove from groups	5 minutes	Automatic
Move to Disabled OU	3 minutes	Automatic
Total per employee	30-45 minutes	Under 5 seconds

2. The Business Problem

In enterprise environments, IT administrators spend significant time on repetitive user provisioning tasks. When HR hires a new employee, IT must manually create an AD account, place it in the correct OU, add it to the right security group, create a home folder, set a temporary password, and communicate credentials to the manager. For offboarding, the process reverses: disable the account, remove group memberships, and archive the user object. In a company hiring 10 people per month, this consumes 5-8 hours of IT time — time that could be spent on infrastructure improvements, security hardening, or automation projects.

This script reduces that to zero manual effort. HR drops a CSV in a folder, and the system handles everything automatically.

3. Production Workflow

The system operates as a folder-watching pipeline that requires no IT intervention for routine onboarding and offboarding:

```
1. HR fills out a CSV with new hires or departing employees
2. HR drops the CSV into C:\Automation\Incoming\ (shared folder)
3. Scheduled task runs hourly, checks for new CSV files
4. Script processes each entry: creates or disables accounts
5. Auto-generated credentials saved to C:\Automation\NewCredentials\
6. Processed CSV archived to C:\Automation\Processed\ with timestamp
7. All actions logged to automation_log.txt for audit trail
```

Folder Structure

```
PS C:\Users\daksh> Get-ChildItem "C:\Automation" -Recurse | Format-Table FullName

FullName
--------
C:\Automation\Incoming
C:\Automation\NewCredentials
C:\Automation\Processed
C:\Automation\automation_log.txt
C:\Automation\employees.csv
C:\Automation\UserAutomation.ps1
C:\Automation\NewCredentials\sofia.martinez.txt
C:\Automation\NewCredentials\tom.bradley.txt
C:\Automation\Processed\2026-04-21_02-46-47_new_hires_april.csv
```

Figure 1: Automation folder structure — Incoming, Processed, NewCredentials, logs

4. Script Architecture

Component	Purpose
OU Mapping	Maps department names to AD Organizational Units
Group Mapping	Maps departments to security groups (GRP-IT, GRP-HR, etc.)
Password Generator	Creates random 12-char passwords with uppercase, lowercase, numbers, special chars
Onboard Function	Creates AD user, adds to group, creates home folder, saves credentials
Offboard Function	Disables account, removes groups, moves to Disabled Users OU
Folder Watcher	Scans Incoming folder for new CSV files
CSV Archiver	Moves processed files to Processed folder with timestamp
Logger	Writes timestamped entries to automation_log.txt

CSV Format

The CSV uses a simple format that HR can fill out without any technical knowledge:

```
Action,FirstName,LastName,Department,JobTitle
Onboard,Tom,Bradley,IT Department,Network Engineer
Onboard,Sofia,Martinez,HR Department,Recruiter
Offboard,James,Wilson,HR Department,
```

5. Onboarding Process

When the script processes an onboard entry, it executes four steps automatically: creates the AD account in the correct OU, adds the user to the department security group, creates a home directory, and generates a credentials file with a random secure password.

Script Execution — Production Mode

```
PS C:\Users\daksh> powershell -ExecutionPolicy Bypass -File "C:\Automation\UserAutomation.ps1"
[2026-04-21 02:46:45] [INFO]
[2026-04-21 02:46:45] [INFO] ##############################################
[2026-04-21 02:46:45] [INFO] #  AUTOMATED USER ONBOARDING AND OFFBOARDING   #
[2026-04-21 02:46:45] [INFO] #  Production Mode - Folder Watcher            #
[2026-04-21 02:46:45] [INFO] #  Script by Daksh Patel                       #
[2026-04-21 02:46:45] [INFO] #  Domain: corp.local                          #
[2026-04-21 02:46:45] [INFO] ##############################################
[2026-04-21 02:46:45] [INFO]
[2026-04-21 02:46:45] [INFO] Processing file: new_hires_april.csv
[2026-04-21 02:46:45] [INFO] Entries: 2 (Onboard: 2 | Offboard: 0)
[2026-04-21 02:46:45] [INFO] ========================================
[2026-04-21 02:46:45] [INFO] ONBOARDING: Tom Bradley (tom.bradley)
[2026-04-21 02:46:47] [SUCCESS] [1/4] AD account created: tom.bradley
[2026-04-21 02:46:47] [SUCCESS] [2/4] Added to group: GRP-IT
[2026-04-21 02:46:47] [SUCCESS] [3/4] Home folder created
[2026-04-21 02:46:47] [SUCCESS] [4/4] Credentials saved to C:\Automation\NewCredentials\tom.bradley.txt
[2026-04-21 02:46:47] [INFO] ONBOARDING COMPLETE for Tom Bradley
[2026-04-21 02:46:47] [INFO] ========================================
Ue8!NEQVZW7T
[2026-04-21 02:46:47] [INFO] ========================================
[2026-04-21 02:46:47] [INFO] ONBOARDING: Sofia Martinez (sofia.martinez)
[2026-04-21 02:46:47] [SUCCESS] [1/4] AD account created: sofia.martinez
[2026-04-21 02:46:47] [SUCCESS] [2/4] Added to group: GRP-HR
[2026-04-21 02:46:47] [SUCCESS] [3/4] Home folder created
[2026-04-21 02:46:47] [SUCCESS] [4/4] Credentials saved to C:\Automation\NewCredentials\sofia.martinez.txt
[2026-04-21 02:46:47] [INFO] ONBOARDING COMPLETE for Sofia Martinez
[2026-04-21 02:46:47] [INFO] ========================================
Ft7!EC5DGAeY
[2026-04-21 02:46:47] [INFO] CSV archived: 2026-04-21_02-46-47_new_hires_april.csv
[2026-04-21 02:46:47] [INFO]
[2026-04-21 02:46:47] [INFO] ##############################################
[2026-04-21 02:46:47] [INFO] #  SUMMARY FOR new_hires_april.csv
[2026-04-21 02:46:47] [INFO] #  Onboarded: 2 | Offboarded: 0
[2026-04-21 02:46:47] [INFO] ##############################################
```

Figure 2: Script processing Tom Bradley and Sofia Martinez — all steps SUCCESS

Auto-Generated Credentials

Each new hire gets a unique randomly generated password. The credentials file is saved for secure delivery to the employee's manager. The user must change their password at first login.

```
PS C:\Users\daksh> Get-Content "C:\Automation\NewCredentials\tom.bradley.txt"
==========================================
  NEW EMPLOYEE CREDENTIALS
  Generated: 2026-04-21 02:46:47
==========================================
  Name:       Tom Bradley
  Username:   tom.bradley
  Password:   Ue8!NEQVZW7T
  Email:      tom.bradley@corp.local
  Department: IT Department
  Title:      Network Engineer
==========================================
  NOTE: User must change password at first login
  Please deliver these credentials securely
==========================================
```

Figure 3: Auto-generated credentials for Tom Bradley — random password, change required at first login

New Users in Active Directory

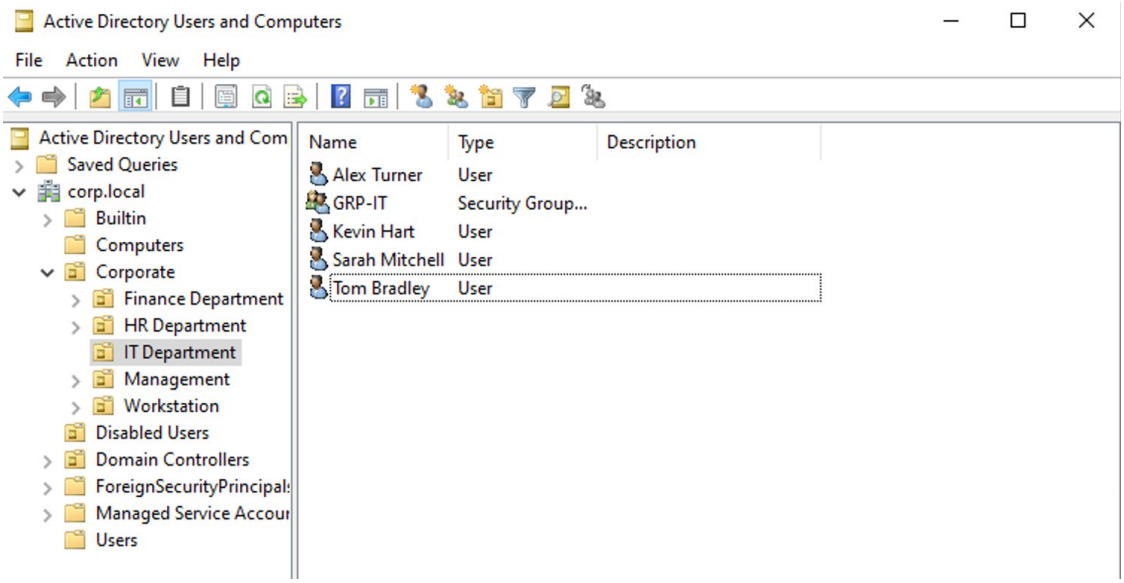

Figure 4: Tom Bradley added to IT Department with GRP-IT group

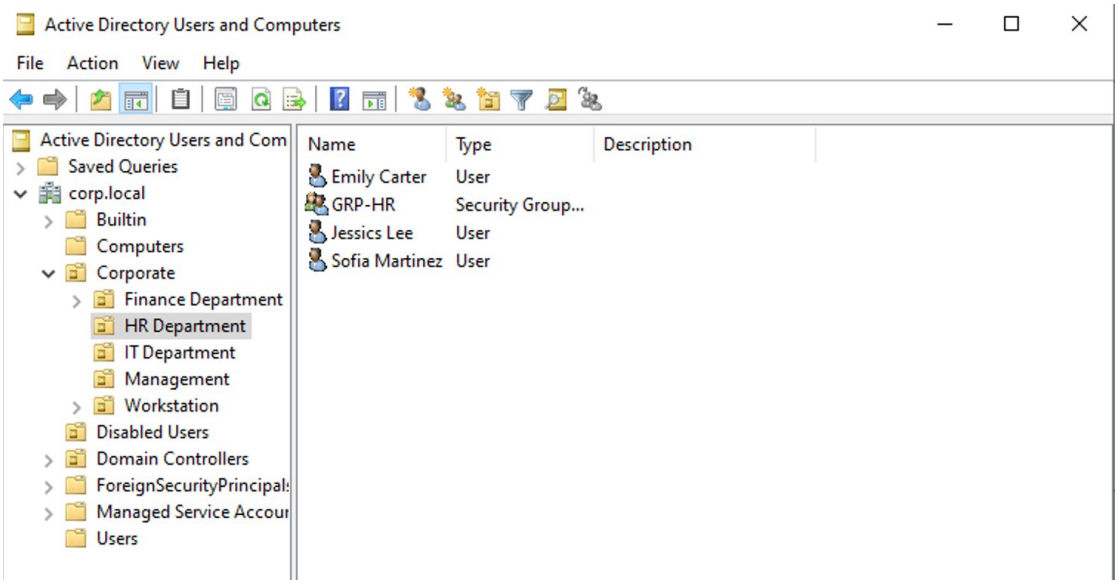

Figure 5: Sofia Martinez added to HR Department with GRP-HR group

6. Offboarding Process

When an employee leaves, the script disables their AD account (preventing login), removes them from all security groups (revoking access to shared folders and resources), and moves the account to a Disabled Users OU for record-keeping and compliance.

Disabled Users — PowerShell Verification

```
PS C:\Users\daksh> Get-ADUser -Filter * -SearchBase "OU=Disabled Users,DC=corp,DC=local" | Format-Table Name,SamAccountName,Enabled

Name          SamAccountName Enabled
----          -------------- -------
James Wilson  james.wilson     False
Lisa Anderson lisa.anderson    False
```

Figure 6: James Wilson and Lisa Anderson — accounts disabled (Enabled: False)

Disabled Users — Active Directory View

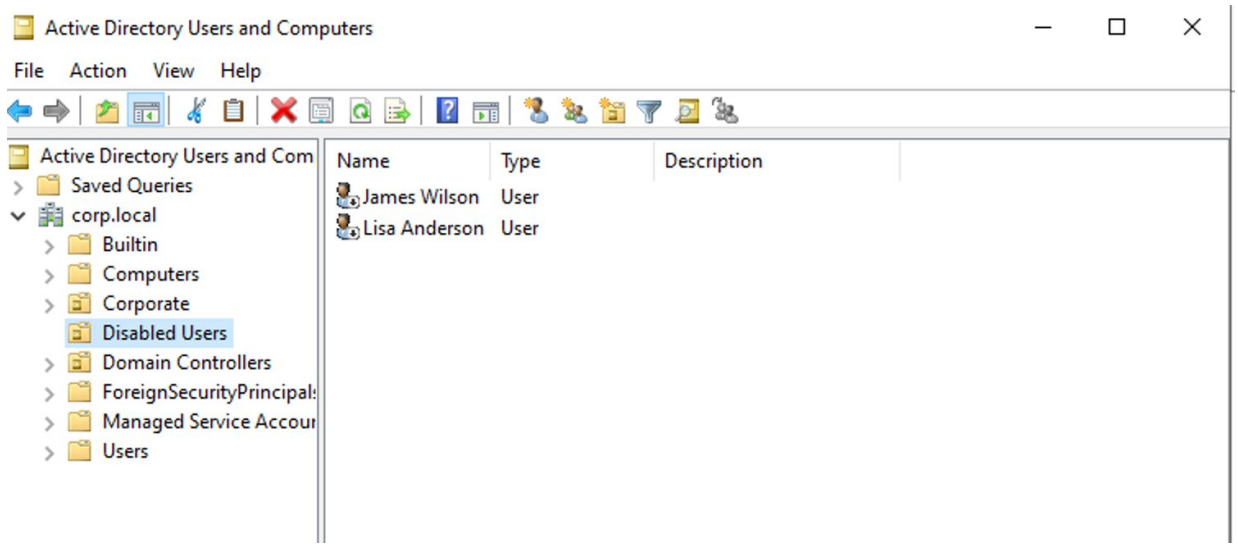

Figure 7: Disabled Users OU in Active Directory — James Wilson and Lisa Anderson moved here

7. Scheduled Automation

A Windows Scheduled Task runs the script every hour, checking the Incoming folder for new CSV files. When HR drops a file, it is processed automatically within the hour — no IT intervention needed. After processing, the CSV is archived with a timestamp to prevent duplicate runs.

Scheduled Task

```
PS C:\Users\daksh> Get-ScheduledTask -TaskName "IT-User-Automation" | Format-List TaskName,State,Description


TaskName    : IT-User-Automation
State       : Ready
Description :
```

Figure 8: IT-User-Automation scheduled task — State: Ready

Archived CSV

```
PS C:\Users\daksh> Get-ChildItem "C:\Automation\Processed"


    Directory: C:\Automation\Processed


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         4/21/2026   2:46 AM            148 2026-04-21_02-46-47_new_hires_april.csv
```

Figure 9: Processed CSV archived with timestamp — prevents re-processing

8. Verification and Testing

All AD Users

```
Name           SamAccountName Department         Enabled
----           -------------- ----------         -------
Alex Turner    alex.turner                          True
Sarah Mitchell sarah.mitchell                       True
Emily Carter   emily.carter                         True
Michael Brown  michael.brown                        True
David Clarke   david.clarke                         True
Rachel Green   rachel.green                         True
Kevin Hart     kevin.hart     IT Department        True
Jessics Lee    jessics.lee    HR Department        True
Ryan Patel     ryan.patel     Finance Department   True
Natasha Kim    natasha.kim    Management           True
Tom Bradley    tom.bradley    IT Department        True
Sofia Martinez sofia.martinez HR Department        True
```

Figure 10: All users in Corporate OU — original 8 + 4 onboarded by script, all Enabled: True

Automation Log

The automation log provides a complete audit trail of every action taken, including timestamps, success/failure status, and details of each operation. This is essential for compliance and troubleshooting.

```
PS C:\Users\daksh> Get-Content "C:\Automation\automation_log.txt"
[2026-04-20 22:09:57] [INFO]
[2026-04-20 22:09:57] [INFO] ##############################################
[2026-04-20 22:09:57] [INFO] #  AUTOMATED USER ONBOARDING AND OFFBOARDING   #
[2026-04-20 22:09:57] [INFO] #  Script by Daksh Patel                       #
[2026-04-20 22:09:57] [INFO] #  Domain: corp.local                             #
[2026-04-20 22:09:57] [INFO] ##############################################
[2026-04-20 22:09:57] [INFO] CSV loaded: 6 entries (Onboard: 4 | Offboard: 2)
[2026-04-20 22:09:57] [INFO]
[2026-04-20 22:09:57] [INFO] ==========================================
[2026-04-20 22:09:57] [INFO] ONBOARDING: Kevin Hart (kevin.hart)
[2026-04-20 22:10:00] [SUCCESS] [1/4] AD account created: kevin.hart in IT Department
[2026-04-20 22:10:00] [SUCCESS] [2/4] Added to group: GRP-IT
[2026-04-20 22:10:00] [SUCCESS] [3/4] Home folder created: C:\Users\HomeDirectories\kevin.hart
[2026-04-20 22:10:00] [INFO] [4/4] ONBOARDING COMPLETE for Kevin Hart
[2026-04-20 22:10:00] [INFO]   Username: kevin.hart | Email: kevin.hart@corp.local | Dept: IT Department | Group: GRP-IT
[2026-04-20 22:10:00] [INFO] ==========================================
[2026-04-20 22:10:00] [INFO] ==========================================
[2026-04-20 22:10:00] [INFO] ONBOARDING: Jessics Lee (jessics.lee)
[2026-04-20 22:10:00] [SUCCESS] [1/4] AD account created: jessics.lee in HR Department
[2026-04-20 22:10:00] [SUCCESS] [2/4] Added to group: GRP-HR
[2026-04-20 22:10:00] [SUCCESS] [3/4] Home folder created: C:\Users\HomeDirectories\jessics.lee
[2026-04-20 22:10:00] [INFO] [4/4] ONBOARDING COMPLETE for Jessics Lee
[2026-04-20 22:10:00] [INFO]   Username: jessics.lee | Email: jessics.lee@corp.local | Dept: HR Department | Group: GRP-HR
[2026-04-20 22:10:00] [INFO] ==========================================
[2026-04-20 22:10:00] [INFO] ==========================================
[2026-04-20 22:10:00] [INFO] ONBOARDING: Ryan Patel (ryan.patel)
[2026-04-20 22:10:00] [SUCCESS] [1/4] AD account created: ryan.patel in Finance Department
[2026-04-20 22:10:00] [SUCCESS] [2/4] Added to group: GRP-Finance
[2026-04-20 22:10:00] [SUCCESS] [3/4] Home folder created: C:\Users\HomeDirectories\ryan.patel
[2026-04-20 22:10:00] [INFO] [4/4] ONBOARDING COMPLETE for Ryan Patel
[2026-04-20 22:10:00] [INFO]   Username: ryan.patel | Email: ryan.patel@corp.local | Dept: Finance Department | Group: GRP-Finance
[2026-04-20 22:10:00] [INFO] ==========================================
[2026-04-20 22:10:01] [INFO] ==========================================
[2026-04-20 22:10:01] [INFO] ONBOARDING: Natasha Kim (natasha.kim)
[2026-04-20 22:10:01] [SUCCESS] [1/4] AD account created: natasha.kim in Management
[2026-04-20 22:10:01] [SUCCESS] [2/4] Added to group: GRP-Management
[2026-04-20 22:10:01] [SUCCESS] [3/4] Home folder created: C:\Users\HomeDirectories\natasha.kim
[2026-04-20 22:10:01] [INFO] [4/4] ONBOARDING COMPLETE for Natasha Kim
[2026-04-20 22:10:01] [INFO]   Username: natasha.kim | Email: natasha.kim@corp.local | Dept: Management | Group: GRP-Management
[2026-04-20 22:10:01] [INFO] ==========================================
[2026-04-20 22:10:01] [INFO] ==========================================
[2026-04-20 22:10:01] [INFO] OFFBOARDING: James Wilson (james.wilson)
[2026-04-20 22:10:01] [SUCCESS] [1/3] Account DISABLED: james.wilson
[2026-04-20 22:10:04] [SUCCESS] [2/3] Removed from group: GRP-HR
[2026-04-20 22:10:04] [INFO] Created Disabled Users OU
[2026-04-20 22:10:04] [SUCCESS] [3/3] Moved to Disabled Users OU
[2026-04-20 22:10:04] [INFO] OFFBOARDING COMPLETE for James Wilson
```

Figure 11: Full automation log — onboarding, offboarding, group operations, all timestamped

Results Summary

Operation	User	Department	Result
Onboard	Kevin Hart	IT Department	SUCCESS — account, GRP-IT, home folder
Onboard	Jessica Lee	HR Department	SUCCESS — account, GRP-HR, home folder
Onboard	Ryan Patel	Finance Department	SUCCESS — account, GRP-Finance, home folder
Onboard	Natasha Kim	Management	SUCCESS — account, GRP-Management, home folder
Onboard	Tom Bradley	IT Department	SUCCESS — auto-password, credentials saved
Onboard	Sofia Martinez	HR Department	SUCCESS — auto-password, credentials saved
Offboard	James Wilson	HR Department	SUCCESS — disabled, removed from GRP-HR, moved to Disabled Users
Offboard	Lisa Anderson	Finance Department	SUCCESS — disabled, removed from GRP-Finance, moved to Disabled Users

9. Lessons Learned

CSV-Driven Automation is HR-Friendly: Using a simple CSV format means HR staff can submit onboarding requests without any technical knowledge. They fill in name, department, and title — the script handles everything else including password generation, OU placement, and group membership.

Idempotent Design Prevents Duplicates: The script checks if a user already exists before creating them. Running it twice with the same CSV produces no errors and no duplicate accounts. This is critical for scheduled automation where the same file might be processed unintentionally.

Offboarding is as Important as Onboarding: Disabling accounts immediately when an employee leaves prevents unauthorized access. Moving them to a Disabled Users OU rather than deleting preserves the account for compliance and legal requirements while ensuring no active access remains.

Auto-Generated Passwords Eliminate Security Risks: Hardcoded passwords in scripts are a security vulnerability. The production version generates random 12-character passwords with mixed case, numbers, and special characters, and requires a change at first login.

Folder Watching Creates a Self-Service Pipeline: The Incoming/Processed folder pattern turns user provisioning into a self-service workflow. HR drops a file, the system processes it, archives it, and logs the result. IT only gets involved for exceptions.

Logging Enables Compliance: Every action is timestamped and logged. This audit trail satisfies compliance requirements and makes troubleshooting straightforward when questions arise about when an account was created or disabled.

End of Documentation

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